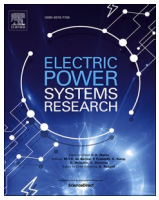
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Forecasting load consumption: a comprehensive evaluation of deep learning and machine learning techniques

Hmeda Musbah [a,*], Meftah Elsaraiti [b]

[a] Libyan Center for Plasma Research, Libya
[b] Higher Institute of Technical Sciences, Syria

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ABSTRACT

Accurate electricity load forecasting is essential for ensuring power grid stability and supporting efficient energy planning. This study presents a comparative analysis of several deep learning models—including Gated Recurrent Unit (GRU), Recurrent Neural Network (RNN), Temporal Convolutional Network (TCN), and Long Short-Term Memory (LSTM)—alongside machine learning models such as CatBoost, XGBoost, and LightGBM. These models were evaluated across six diverse hourly load consumption datasets from Portugal, Nova Scotia, France, Italy, Denmark, and Spain, covering one-year and three-year periods. Randomized Grid Search was used to fine-tune model hyperparameters. The results show that model performance varies by region and dataset duration. LightGBM consistently outperformed other models, achieving the best performance on the Spain dataset with an nRMSE of 2.31 %, MAE of 397.8, and R^2 of 0.98 for the one-year dataset, and an improved nRMSE of 1.3 % for the three-year dataset. XGBoost also performed well with an nRMSE of 3.09 % and R^2 of 0.98 on the Italy dataset. In contrast, LSTM showed weaker results with an nRMSE of 9.53 % and R^2 of 0.60 on the Portugal dataset. Among deep models, RNN stood out slightly, achieving an nRMSE of 4.8 % and R^2 of 0.90 on the Spain dataset. Overall, gradient boosting models proved to be more robust and accurate for short-term load forecasting tasks.

1. Introduction

Electrical companies always seek to generate sufficient power to meet load consumption. Estimating and forecasting load consumption helps these companies in the planning and decision-making process [1]. Accurate forecasting ensures a balance between electricity generation and demand, which is essential for reducing operational costs and preventing outages. However, the increasing complexity of power grids, along with the integration of renewable energy sources and evolving consumption patterns, makes load forecasting challenging. Load forecasting involves predicting very short, short, medium, or long-term load values using historical datasets [2,3]. Several statistical models, including Autoregressive (AR), Moving Average (MA), and Autoregressive Integrated Moving Average (ARIMA) models [4,5], have been used to achieve reliable forecasting. One problem with statistical models is that data is subject to fixed assumptions such as data stationarity and distribution, since violations of these assumptions can cause inaccurate forecasting. To address non-stationarity in data, [6] proposed a technique called the adaptive DC technique. After evaluating the performance of this technique on several datasets, researchers found that it was able to eliminate stationarity in one step.

Another category of load forecasting models used to improve accuracy is artificial intelligence models. These models are designed to mimic the human brain by learning from past experiences and producing generalized results. Deep learning models have the ability to capture complex patterns or non-linear relationships in the data [7]. Therefore, this feature attracted researchers to employ these models in load forecasting. However, current research does not provide a thorough comparison of machine learning models for predicting wind speed and power over different time frames. In addition, tree-based approaches such as scalable boosting machines and random forests have received limited attention.

Despite the growing popularity of deep learning techniques, they often require large amounts of data and are sensitive to noise and outliers. In addition to deep learning models, machine learning models such as CatBoost, XGBoost and LightGBM and support vector machines [3] have demonstrated their ability to achieve high accuracy. These models are generally more robust to smaller datasets and often deliver competitive results with lower computational costs. However, there is a lack of comprehensive studies comparing these methods under uniform

* Corresponding author.
E-mail address: Hm392855@dal.ca (H. Musbah).

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settings across diverse regions and data conditions.

This study addresses this gap by conducting a detailed comparison between selected deep learning and machine learning models across six diverse load consumption datasets, including datasets from Portugal, Nova Scotia, France, Italy, Denmark, and Spain. The evaluation considers various data characteristics, dataset sizes, and model configurations to identify the most effective forecasting approaches under different scenarios.

The main contributions of this work can be summarized as follows:

1. The Randomized Grid Search technique is employed to optimize the hyperparameters of the models, enhancing their predictive capabilities.
2. Examination of the effect of dataset size on the performance of the models, where the dataset size increased from one year to three years.
3. Investigation of the impact of dataset characteristics, including outliers, noise, and randomness, on the performance of the forecasting models, contributing to a deeper understanding of the factors influencing load forecasting accuracy and reliability.
4. Through this comparative study, insights into the strengths and weaknesses of each model are gained, aiding in the selection of the most suitable approach for load forecasting applications.

The remainder of this paper is organized as follows: Section 2 presents previous work related to this study. Section 3 describes the study's methodology, including the framework, data description, methods, evaluation metrics, and hyperparameter tuning. Section 4 presents the results and discussion. Finally, Section 5 presents the conclusions of this work.

2. Literature review

Efforts to optimize electricity consumption are vital in the world today context. In the field of energy conservation research, the prediction of electrical loads emerges as a method possessing significant practical and scientific value. This is especially pertinent given the dynamic accuracy standards inherent in the wholesale electricity and energy market [8]. Numerous scholarly articles underscore the efficacy of energy conservation achieved through forecasting power consumption [9–11]. Researchers employ diverse techniques for power consumption forecasting. These include classical and deep machine learning methods [12–14], mathematical models incorporating fuzzy logic [15], and models that account for seasonal patterns in power consumption time series [16], among other methodologies [17]. Artificial neural networks have garnered significant attention in this space, particularly Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), and Recurrent Neural Networks (RNN) [18,19]. Additionally, newer machine learning models such as CatBoost [20], Light Gradient Boosting Machine (LightGBM) [21], and Temporal Convolutional Networks (TCN) [22] have also demonstrated efficacy in forecasting tasks.

LSTM models, a specialized type of RNN, have proven particularly effective in handling the vanishing and exploding gradient problems common in training deep networks, especially for long-term time series forecasting [23]. Studies on short-term electricity consumption prediction further validate the effectiveness of LSTM and other machine learning methods, including random forest algorithms, XGBoost, and CatBoost [24]. GRU-based methods have also been proposed for short-term load prediction, offering improved performance and efficiency compared to traditional techniques [25]. In industrial settings, machine learning models have been tailored to forecast electricity consumption, with CatBoost emerging as a key tool due to its high accuracy and reliability [26].

While RNNs are effective for short-term forecasting, their limitations with long-term dependencies are well noted. To address this, LSTM models are often used for long-range predictions due to their specialized architecture [27]. Some studies have even integrated genetic algorithms with LSTM models to further refine forecasting accuracy [28]. The LSTM's effectiveness is particularly evident in household load forecasting, where it captures meaningful temporal relationships despite the inherent variability in consumption patterns. This capability is enhanced when predicting aggregated household loads and accounting for intra-house components [29].

Temporal Convolutional Networks (TCNs) have also been applied for short-term load forecasting. These models have shown superior accuracy in handling nonlinear time series data and outperform traditional methods in extracting key features [30]. Comparative studies between TCN, GRU, and LSTM suggest that some newer algorithms may require less computational time, particularly with high-frequency or high-dimensional inputs [31]. Additionally, combining GRU for temporal feature extraction with LightGBM for final forecasting has shown to improve model performance by up to 2 % across several metrics [32].

Beyond deep learning, gradient boosting techniques such as XGBoost and LightGBM have achieved strong results in real-world applications. For example, XGBoost successfully forecasted 24-hour ahead electricity usage using half-hourly data from the Australian System Operator [33]. LightGBM also demonstrated strong performance in industrial contexts in China and Ireland, outperforming both SVM and LSTM models [34]. Similarly, CatBoost has been proposed for load forecasting in technical industrial zones, achieving better accuracy than LSTM-based counterparts [35].

Studies comparing multiple ANN architectures—including MLP, CNN, LSTM, and GRU—showed that GRU generally resulted in the lowest prediction error, followed by LSTM and CNN respectively [36]. Beyond the energy domain, TCNs have also shown promising results in weather forecasting tasks, where they outperformed LSTM models in various time series applications [37].

Delayed data was used to improve energy consumption prediction. Five deep learning models were tested. Augmenting the data and delayed energy data significantly improved accuracy. LSTM and BiGRU models achieved the best performance, outperforming existing methods and supporting efficient energy management [38]. An LSTM-based deep learning method was used to accurately predict short-term solar photovoltaic (PV) generation, leveraging LSTM's ability to model time series relationships [39]. A graph neural network-based approach for spatiotemporal prediction of photovoltaic power generation has been proposed, modeling photovoltaic systems as graph nodes to capture spatial and temporal dependencies [40]. The analyzes LSTM-RNN effectiveness in predicting low-flow time series and compare it with RNN and naïve method [41].

This study investigates the use of LSTM-RNN for low-flow time series forecasting using daily discharge data from the Basantapur gauging station in India. Various performance metrics, including RMSE, ENS, correlation coefficient, and MAE, were used for evaluation. Results show that LSTM-RNN outperformed both the RNN and naïve methods in accuracy and efficiency. The study concludes that LSTM-RNN is a reliable AI technique for low-flow forecasting.

Despite the extensive research on load consumption forecasting using ML and DL models, several key gaps remain unaddressed. Below, we highlight the main limitations in existing studies that motivate our work.

1. Lack of comparative studies: While existing research has explored both machine learning (ML) and deep learning (DL) models for load consumption forecasting, comprehensive comparisons across multiple datasets remain limited.
2. Regional performance variations: Previous studies often focus on specific datasets rather than examining how different models perform across diverse geographical regions.
3. Impact of dataset length: The effect of different time spans (one-year vs. three-year datasets) on model performance has not been extensively analyzed.

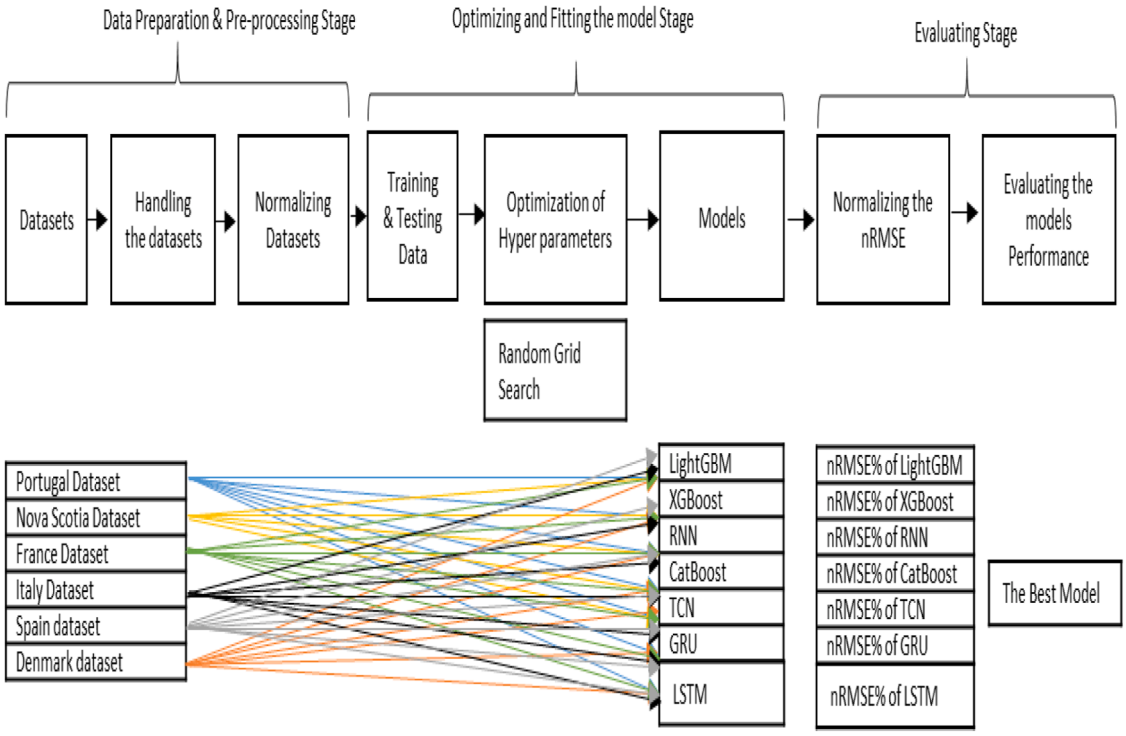


Fig. 1. The methodology for comparing various deep learning and machine learning models.

Table 1
Statistical summary of six datasets.

Statistic	Denmark	France	Italy	Nova Scotia	Portugal	Spain
Min	2231.00	30,426.00	17,636.00	8126.68	3393.00	18,616
Max	5710.00	90,588.00	55,157.00	23,916.36	8578.00	40,324
25 %	3108.00	44,761.25	26,321.25	11,164.91	4763.00	24,492
Median	3709.00	51,981.50	32,610.50	12,761.14	5607.50	28,664
75 %	4375.00	61,311.75	39,361.00	14,513.24	6287.25	31,719
Mean	3745.38	53,572.74	33,053.67	12,987.79	5596.56	28,420
Std	760.14	11,553.40	7627.10	2559.75	966.06	4644.4

3. Methodology

The methodology for comparing various deep learning and machine learning models for load forecasting is divided into three stages, as shown in Fig. 1. These stages are:

3.1. Dataset description stage

Many load consumption datasets from various regions, including Portugal, Nova Scotia, France, Italy, Denmark, and Spain, were used in this study. The six datasets were carefully selected to represent a diverse range of geographic locations, climates, and energy consumption behaviors. Specifically, the datasets include regions from Southern Europe (Portugal, Spain, Italy), Northern and Western Europe (France, Denmark), and North America (Nova Scotia). This selection allows for the analysis of model performance under varying conditions such as seasonal load variability, population density, and energy usage patterns. The inclusion of these datasets was intended to evaluate the generalizability of the forecasting models and ensure the robustness of the results across different consumption environments. These datasets contain one variable recorded hourly over the course of one year, from Jan 1, 2015, to Dec 31, 2015.

In the pre-processing step, missing data and outliers were identified and addressed. Since the datasets come from different regions, their maximum and minimum values vary, as shown in Table 1. The range of variation between the maximum and minimum values was normalized to a scale between 0 and 1 using Eq. (16).

$$X_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}} \tag{16}$$

Where: x represents the original data point or variable. x_min is the minimum value of the original data set. x_max is the maximum value of the original data set.

Thus, X_norm represents the normalized version of the input data set x.

The prepared dataset is divided into two portions: an 80 % training dataset and a 20 % testing dataset. The training dataset is used to train the model, while the testing dataset is used as an unseen dataset.

3.1.1. Data analysis

The statistical summary table provides essential insights into the distribution and range of values across the six datasets, Table 1. One of the most striking observations is the wide variability in value ranges—France and Spain have significantly higher means and maximum values compared to datasets like Denmark and Portugal. For instance, France has a maximum value of 90,588 and a mean over 53,000, while Denmark's max is just 5710 with a mean of around 3745. This kind of scale difference can heavily influence model behavior if not properly normalized or scaled, potentially leading to biased predictions or convergence issues, especially in models sensitive to magnitude

differences like neural networks. The standard deviations also tell us about the spread of the data. France and Italy exhibit high standard deviations (11,553 and 7627 respectively), indicating more variability and a broader distribution. In contrast, Denmark and Portugal have relatively lower variability, suggesting more tightly clustered values. This could make it easier for models to learn from those datasets but might also mean they're less generalizable if patterns are too narrow. The interquartile range (the difference between the 25 % and 75 % values) shows us the core concentration of data. For example, Portugal's IQR is much smaller than France's, reflecting a more consistent dataset that could be easier for a model to predict accurately. On the other hand, the Nova Scotia dataset shows considerable spread (e.g., a wide range between 25 % and 75 %), which might lead to higher prediction errors if the model doesn't handle variability well.Overall, these differences in statistical characteristics highlight why preprocessing (like normalization) is crucial. From a modeling perspective, datasets with large ranges or variability may lead to higher nRMSE or MAE values unless properly handled. This is exactly why nRMSE (normalized root mean squared error) becomes a preferred evaluation metric: it adjusts for the scale of the target variable, allowing for a fair comparison across datasets with vastly different value ranges—something that raw error metrics like MAE or MAPE cannot always capture effectively.

3.2. Methods

In this stage, several models were selected based on their proven effectiveness in time series forecasting, particularly in energy and load prediction tasks. These models include both machine learning and deep learning techniques, chosen for their ability to capture complex patterns and temporal dependencies in the data. The selection process aimed to evaluate the performance of each model across various datasets to determine the most suitable approach for accurate forecasting.

- CatBoost Algorithm

CatBoost is a supervised machine learning algorithm based on gradient boosting, designed to handle categorical data efficiently. Unlike traditional algorithms that require converting categorical variables into numerical values, CatBoost processes them directly, reducing data preparation effort and improving model performance. It uses decision trees as weak learners to iteratively minimize errors and build accurate predictive models [42].

- LightGBM algorithm

LightGBM is a gradient boosting framework that uses a histogram-based algorithm, which speeds up training by converting continuous feature values into discrete bins. This approach reduces memory usage and enables faster computation. LightGBM grows trees using a leaf-wise split strategy, creating more complex trees and achieving higher accuracy compared to level-wise methods. It performs well on large datasets with faster training times than XGBoost. However, its leaf-wise growth can lead to overfitting, especially on small datasets, making it sensitive to overfitting issues [43].

- The XGBoost algorithm

XGBoost is an advanced machine learning algorithm based on Gradient Boosting Decision Trees (GBDT), designed for high performance and accuracy, especially with structured data. It builds a strong model by combining multiple weak classifiers (decision trees) and improving them iteratively based on previous errors. XGBoost handles missing values and outliers efficiently, includes built-in regularization to prevent overfitting, and scales well to large datasets [43].

- Temporal Convolutional Networks (TCNs)

Temporal Convolutional Networks (TCNs) are deep learning models designed for sequence and time-series data tasks like forecasting, anomaly detection, and sequence classification. They use convolutional layers with dilation to capture long-range dependencies efficiently. TCNs feature an Encoder Decoder architecture and address the gradient decomposition problem by combining outputs from blocks with different dilation rates. Compared to traditional RNNs and LSTMs, TCNs offer improved performance, faster training, and better handling of sequence data [44].

- Recurrent Neural Networks (RNN)

Recurrent Neural Networks (RNNs) are powerful architectures designed to model time-dependent and sequential data problems, such as stock market prediction, artificial intelligence applications, and text generation. RNNs are particularly effective because they maintain a memory of previous inputs, allowing them to capture temporal dependencies in data. However, despite their potential, RNNs face significant challenges during training, primarily due to the vanishing gradient problem. Gradients, which carry important information about error signals used for parameter updates, can become extremely small as they are propagated back through many time steps. When the gradients shrink to near-zero values, the parameters of network hardly update, making it difficult for the RNN to learn long-term dependencies in data sequences [28].

- Long Short Term Memory (LSTM)

Long Short Term Memory (LSTM) is a type of Recurrent Neural Network (RNN) designed to overcome the vanishing gradient problem in traditional RNNs. LSTMs are effective in handling long range dependencies and variable length sequences, making them suitable for time series analysis. Their architecture includes memory cells and gating mechanisms that control information flow and maintain important data over time. However, LSTMs come with challenges such as high computational complexity, risk of overfitting, long training times, and limited interpretability. Additionally, tuning hyperparameters for optimal performance can be a difficult and time consuming process [28].

- Gated recurrent unit (GRU)

Gated Recurrent Unit (GRU) is a type of Recurrent Neural Network (RNN) designed as a simpler alternative to Long Short Term Memory (LSTM) networks. GRUs merge the forget and input gates into a single update gate and combine the cell state with the hidden state, reducing complexity and the number of parameters. GRUs are faster, require less memory, and are easier to train than LSTMs, making them suitable for tasks with limited computational resources or shorter sequences. However, LSTMs may perform better on datasets with longer sequences due to their more complex memory management. GRU offers efficient performance for modeling sequential data while maintaining competitive accuracy in many applications [45].

3.3. Optimizing and fitting the model stage

The Random Grid Search technique is used to optimize the hyper-parameters of the machine and deep learning models. Essentially, the technique involves exhaustively searching through a manually specified range of the hyper-parameter space, thereby avoiding the computational burden of deeply evaluating all possible combinations. The combination of hyper-parameters that leads to better results is chosen to be the final set of hyper-parameters. The main disadvantage of the Random Grid Search is that it is difficult to reach the optimal global solution if the search range is limited. The tuning process involved parameters such as learning rate, number of estimators, maximum depth, and subsample ratios. We selected the best model based on validation performance

Table 2
The optimized hyper-parameters of the models.

The model	The hyper-parameter
LightGBM	N estimators is [100, 500, 1000]
XGBoost	Subsample is [0.5, 0.75, 1.0]
	Learning rate is [0.01, 0.1, 0.3]
	max_depth is [3, 6, 9]
CatBoost	100 between and 1000 for number of trees in the forest
	[0.01, 0.05, 0.1, 0.2] for gradient boosting Learning rate
	(3,5,10, 15) Depth of the trees Depth
TCN	Epoch is between 50 and 150
	num_channels is [64, 64, 64], [64, 128, 256]
	kernel_size is (2, 5)
	dropout is [0.2, 0.3, 0.4]
GRU	Random integer between 50 and 150 for number of epochs.
RNN	Random integer between 50 and 200 for hidden layer size.
LSTM	Random integer between 1 and 3 for number of layers.

metrics such as RMSE and R^2. Table 2 shows the hyper-parameters that were optimized and their search range in this study and their search range.

3.4. Evaluating stage

In this paper, the assessment metrics utilized include mean absolute error (MAE), mean absolute percentage error (MAPE), and root mean square error (RMSE). MAE addresses the issue of mutual cancellation during error calculations, providing an accurate representation of the actual prediction error magnitude. MAPE is frequently employed to measure prediction accuracy, with a value closer to 0 % indicating a smaller prediction error. RMSE, derived from the root mean square error percentage, offers a better characterization of the fit between measured and predicted values. A smaller Root Mean Squared Error (RMSE) value corresponds to higher prediction accuracy [46–48],. The metrics are expressed by the following equations [3]:

$$MAE = \frac{1}{N}\sum_{t=1}^{N}\left|y_a - y_f\right| \tag{13}$$

$$MAPE = \frac{1}{N}\sum_{t=1}^{N}\left|\frac{y_a - y_f}{y_a}\right| \tag{14}$$

$$RMSE = \sqrt{\frac{\sum_{t=1}^{N}\left(y_a - y_f\right)^2}{N}} \tag{15}$$

Where N is the number of errors, y_a is the actual value and y_f is the forecast value.

4. Results and discussion

Table 3 showcases how well the LightGBM model performs across six different datasets, consistently delivering accurate results with low error rates and strong R^2 values. Among all, the Spain dataset emerges as the top performer, showing the lowest nRMSE % of 2.31, MAE of 397.8, and MAPE % of 1.4, alongside an impressive R^2 of 0.98. Despite the wide data range from 18,616 to 40,324, the model with 500 estimators, a 0.1 learning rate, and depth 6, effectively captured complex patterns without overfitting. Portugal follows closely with a nRMSE % of 2.57 and the lowest MAE of 95, likely helped by its relatively narrow value range from 3393 to 8578 and similar hyperparameters. On the other end, Nova Scotia recorded the highest nRMSE % of 3.47, which, despite a high R^2 of 0.98, suggests the model struggled with its unique distribution and broad value scale from 8126.68 to 23,916.36. France, Italy, and Denmark fall in the middle range, with France showing a slightly weaker R^2 of 0.94 and higher MAPE %, which may reflect greater sensitivity to outliers in its large-scale data from 30,426 to 90,588, even with deeper trees and a higher learning rate. After weighing all performance indicators and accounting for data scale and model complexity, the rankings are: 1) Spain, 2) Portugal, 3) Italy, 4) Denmark, 5) France, 6) Nova Scotia.

Table 4 displays the performance of the XGBoost model, which maintains strong and consistent accuracy across a variety of datasets despite differences in scale and complexity. Among all, Spain stands out as the top performer with the lowest nRMSE % of 2.36, relatively low MAE of 501.9, and MAPE % of 1.77, supported by a solid R^2 score of 0.97. This is especially notable considering Spain used a deep model with depth of 9 and a very low learning rate of 0.01, which likely contributed to its precise learning across a wide data range from 18,616 to 40,324. Portugal also performed impressively, achieving an nRMSE % of 2.74 and the lowest MAE of 115, likely aided by a narrower data range and a balanced setup of 500 estimators and a 0.3 learning rate. In contrast, France showed the weakest performance, with the highest nRMSE % of 3.86, highest MAPE % of 4.84, and lowest R^2 of 0.832, indicating the model struggled to generalize well despite similar hyperparameter settings as other datasets. Nova Scotia, Italy, and Denmark showed fair to good results, with Nova Scotia slightly ahead thanks to a competitive nRMSE % of 2.93 and an R^2 of 0.97, although its MAE of 233.4 reflects the relatively larger scale of values from 8126.68 to 23,916.36. Ranking the overall performance of the XGBoost model

Table 3
The hyper-parameters of the LightGBM model.

Dataset	N estimators	learning rate	depth	nRMSE %	MAE	MAPE %	R^2
Portugal dataset	500	0.1	6	2.57	95	1.71	0.97
Nova Scotia dataset	100	0.1	3	3.47	182.7	1.55	0.98
France dataset	100	0.3	9	3.15	138.3	2.5	0.94
Italy dataset	500	0.1	6	2.92	615	1.8	0.98
Denmark dataset	100	0.3	9	3.28	92.6	2.3	0.97
Spain dataset	500	0.1	6	2.31	397.8	1.4	0.98

Table 4
The Hyper-parameters of The XGBoost Model.

Dataset	N estimators	learning rate	depth	nRMSE %	MAE	MAPE %	R^2
Portugal dataset	500	0.3	3	2.74	115	2.04	0.966
Nova Scotia dataset	100	0.1	3	2.929	233.4	1.99	0.97
France dataset	100	0.1	3	3.862401	2661	4.84	0.832
Italy dataset	500	0.3	3	3.09	702.39	2.07	0.98
Denmark dataset	100	0.1	3	3.44			
Spain dataset	500	0.01	9	2.36	501.9	1.77	0.97

Table 5
The hyper-parameter values of CatBoost model.

Data set	iterations	learning rate	depth	nRMSE %	MAE	MAPE %	R^2
Portugal dataset	566	0.1	4	2.53	95	1.69	0.97
Nova Scotia dataset	761	0.01	5	3.22	702.39	2.07	0.98
France dataset	443	0.2	4	2.85	1373	2.5	0.95
Italy dataset	558	0.2	5	2.78	604	1.81	0.98
Denmark dataset	558	0.2	5	2.91	77.46	1.99	0.98
Spain dataset	199	0.2	7	2.3	407.9	1.46	0.98

Table 6
The hyper-parameter values of TCN model.

Dataset	kernel size	epoch	dropout	num channels	nRMSE %	MAE	MAPE %	R^2
Portugal dataset	3	100	0.4	[64, 128, 256]	6.2	360.2	6.5	0.7
Nova Scotia dataset	4	100	0.4	[64, 64, 64	6.64	712.4	6.42	0.81
France dataset	3	100	0.2	[64, 64, 64]	8.39	4214	7.36	0.57
Italy dataset	4	100	0.4	[64, 64, 64]}	7.51	2374.18	7.4	0.83
Denmark dataset	4	100	0.3	[64, 128, 256]	7.18	244.94	6.11	0.81
Spain dataset	4	100	0.4	[64, 64, 64]	6.63	1933	7.16	0.72

Table 7
The hyper-parameter values of GRU model.

Dataset	epochs	Hidden layer size	Num layers	nRMSE %	MAE	MAPE %	R^2
Portugal dataset	138	98	1	7.99	310.76	5.69	0.83
Nova Scotia dataset	125	107	2	7.10	839.41	7.7	0.7
France dataset	132	124	1	7.47	290	6.59	0.72
Italy dataset	138	98	1	9.88	284	9.33	0.78
Denmark dataset	137	166	2	9.10	243	6	0.83
Spain dataset	138	98	1	6.29	305.4	5.1	0.82

across datasets: 1) Spain, 2) Portugal, 3) Nova Scotia, 4) Italy, 5) Denmark, 6) France.

Table 5 presents the CatBoost model's performance across various datasets, where it consistently demonstrated high accuracy with only subtle variations influenced by dataset scale and hyperparameters. Among all, Spain achieved the most impressive results, posting the lowest nRMSE % of 2.3, low MAE of 407.9, and the best MAPE % of 1.46, coupled with a strong R^2 of 0.98. This performance is particularly notable considering the smaller number of iterations, 199, and deeper trees with depth of 7, which appear to have helped capture more complex patterns in Spain's mid-range data from 18,616 to 40,324. Portugal also showed excellent performance, with a nRMSE % of 2.53 and an impressively low MAE of 95, making it particularly accurate in smaller-scale data ranges from 3393 to 8578. Italy and France followed closely, both achieving nRMSE % under 2.9, with Italy having slightly better overall error metrics. Denmark and Nova Scotia performed acceptably, but Denmark's higher nRMSE % of 2.91 and Nova Scotia's notably large MAE of 702.39 reveal some struggle in these datasets, possibly due to the broader numeric range and smaller error tolerance. Ranking the models from best to worst: 1) Spain, 2) Portugal, 3) Italy, 4) France, 5) Denmark, 6) Nova Scotia.

Table 6 shows that the TCN model delivers moderately accurate predictions across the six datasets, with some variations influenced by data scale, sequence complexity, and model configuration. The Portugal dataset had the best performance, with the lowest nRMSE % of 6.2 and reasonable MAE and MAPE % values of 360.2 and 6.5, respectively, although the R^2 score of 0.7 suggests room for improvement in capturing variance. Spain and Nova Scotia follow closely, with nRMSE % values of 6.63 and 6.64, respectively. Despite having similar architectures (kernel size 4, dropout 0.4), Nova Scotia had a much larger data range and a notably higher MAE, reflecting the challenge of scaling predictions in such contexts. France showed the weakest performance, with the highest nRMSE % of 8.39, MAE of 4214, and a low R^2 of 0.57, indicating that the model struggled to generalize well in this high-variance dataset, even with a relatively low dropout rate of 0.2. Italy and Denmark also had higher nRMSE % values (7.51 and 7.18), with Italy showing a high MAPE %, and Denmark delivering a decent MAE but still struggling with variance explanation ($R^2 = 0.81$). Based on the overall model performance and considering both error metrics and data characteristics, the ranking is: 1) Portugal, 2) Spain, 3) Nova Scotia, 4) Denmark, 5) Italy, 6) France.

Table 7 exhibits how the GRU model performs across a variety of datasets with different scales and complexities. Among all, the Spain dataset stands out as the best-performing case, achieving the lowest nRMSE of 6.29 %, a low MAE of 305.4, and the lowest MAPE of 5.1 %, along with a solid R^2 score of 0.82. This is particularly impressive considering the wide value range from 18,616 to 40,324 and a relatively simple architecture with one layer and 98 hidden units. Portugal and

Table 8
The hyper-parameter values of RNN model.

Dataset	epochs	Hidden layer size	Num layers	nRMSE %	MAE	MAPE %	R^2
Portugal dataset	135	26	1	4.9	188	3.4	0.92
Nova Scotia dataset	135	26	1	4.52	518.7	4.48	0.88
France dataset	135	26	1	6.12	2766	4.98	0.81
Italy dataset	135	26	1	6.09	1783	5.45	0.91
Denmark dataset	121	28	1	6.281	178.9	4.42	0.89
Spain dataset	121	28	1	4.8	1028.9	3.73	0.9

Table 9
The hyper-parameter values of LSTM model.

Dataset	epochs	Hidden layer size	Num layers	nRMSE %	MAE	MAPE	R^2
Portugal dataset	161	100	2	9.53	470.6	0.08	0.6
Nova Scotia dataset	163	70	1	7.88	856.2	0.07	0.7
France dataset	163	70	1	9.23	4325	0.07	0.59
Italy dataset	160	70	1	13.51	2815.2	0.08	0.79
Denmark dataset	160	70	1	9.5	324	0.083	0.71
Spain dataset	129	55	2	10.05	1940	0.074	0.71

France also perform relatively well, with Portugal recording a slightly higher nRMSE of 7.99 % but benefiting from a small value range of 3393 to 8578 and a strong R^2 of 0.83. France, while showing a balanced MAPE and a low MAE of 290, still has an nRMSE of 7.47 % and a weaker R^2 of 0.72, likely due to its large-scale data ranging from 30,426 to 90,588, which challenges the model's capacity. In contrast, Italy and Denmark demonstrate the weakest performance. Italy reaches the highest nRMSE of 9.88 % and MAPE of 9.33 %, despite a relatively small MAE, which suggests inconsistent predictions across varying magnitudes. Denmark similarly shows a high nRMSE of 9.10 %, with a low MAE likely due to the dataset's smaller scale, but the model underperforms overall. Nova Scotia, despite using a deeper GRU with two layers and a hidden size of 107, ends up with a higher nRMSE of 7.10 % and the lowest R^2 of 0.70, possibly indicating overfitting or an under-representation of temporal patterns within its narrower data range. Ranking the datasets from best to worst performance based on overall metrics and model fit: 1) Spain, 2) Portugal, 3) France, 4) Nova Scotia, 5) Denmark, 6) Italy.

Table 8 highlights the RNN model's performance across multiple datasets, showing a relatively consistent level of accuracy with modest depth—1 layer—and small hidden sizes ranging from 26 to 28. Among the results, Nova Scotia achieved the best performance in terms of nRMSE, 4.52 percent, followed closely by Spain at 4.8 percent and Portugal at 4.9 percent. Nova Scotia's strong performance is particularly noteworthy given the high numerical scale of the dataset, ranging from 8126 to 23,916, suggesting the model managed to generalize well despite its simple architecture. However, Portugal stands out with the highest R-squared value of 0.92 and the lowest MAPE at 3.4 percent, indicating more stable and reliable predictions relative to the actual values. Spain also performed well across all metrics, delivering low errors and a high R-squared value of 0.9, making it a strong second contender. France, Italy, and Denmark, on the other hand, showed higher nRMSE values above 6 percent and lower R-squared scores, indicating weaker model fits, especially in the case of France, which had an R-squared of 0.81. Despite using the same architecture across datasets, the varying results suggest that RNNs are more sensitive to the scale and complexity of the data than to hyperparameter adjustments. Based on overall performance, the models rank as follows: first, Nova Scotia; second, Portugal; third, Spain; fourth, Denmark; fifth, Italy; and sixth, France. While metrics such as MAE and MAPE provide meaningful local insights, they can distort cross-dataset comparisons due to differing value ranges. By normalizing the error relative to the data scale, nRMSE offers a fair and consistent metric, making it the most effective choice for evaluating performance across datasets with varied numerical domains.

Table 9 shows that the LSTM model generally struggled across all datasets, with higher error rates and lower R^2 scores compared to other models analyzed earlier. Among all, Nova Scotia had the most favorable performance, achieving the lowest nRMSE % of 7.88, a relatively moderate MAE of 856.2, and an acceptable R^2 of 0.70, despite having only one layer and a smaller hidden size of 70. Interestingly, France and Portugal followed closely with nRMSE values of 9.23 and 9.53, respectively, but both had notably low R^2 scores 0.59 and 0.60 respectively, indicating poor model fit and limited generalization. Spain and Denmark, while similar in structure to other datasets, didn't fare much better—Spain showed a high nRMSE % of 10.05, and Denmark had 9.5 with R^2 at just 0.71, despite their relatively simple data scales. Italy, however, clearly underperformed with the highest nRMSE % of 13.51 and MAE of 2815.2, showing that the model failed to capture meaningful temporal trends in this case, despite using the same network structure as better-performing datasets. Ranking the models based on performance: 1) Nova Scotia, 2) France, 3) Portugal, 4) Denmark, 5) Spain, 6) Italy. While metrics like MAE and MAPE (or their decimal counterparts here) provide error magnitudes and relative deviation, nRMSE % offers a normalized view that makes performance comparison across diverse datasets more objective and fair—especially when dealing with varying data scales and ranges. In this case, it clearly highlighted the weaker generalization of LSTM models across all datasets.

While MAE and MAPE offer important localized insight into absolute and relative errors, nRMSE % is the most informative and fair metric

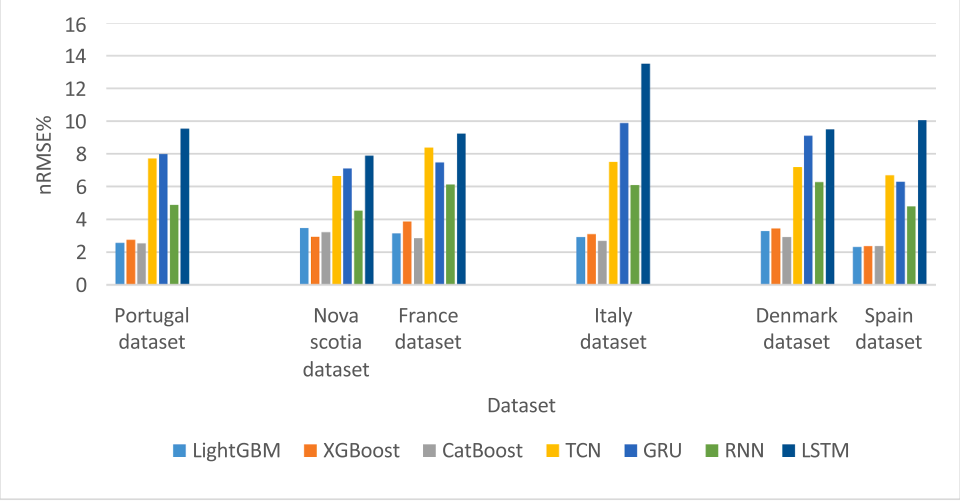


Fig. 2. summarizes the accuracy of the models across the datasets.

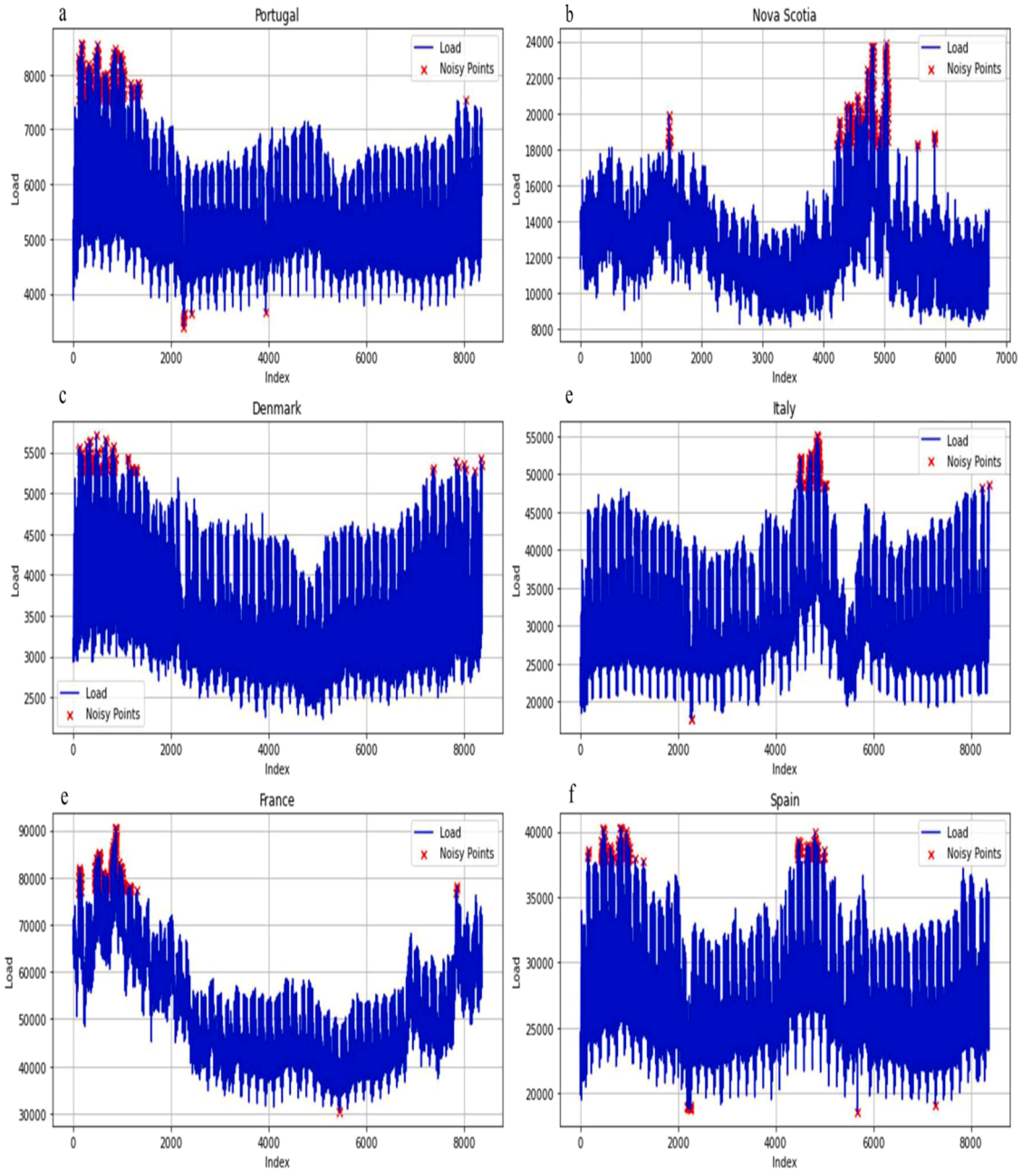


Fig. 3. illustrates the distribution of noisy points in the datasets. (a) Portugal: Moderate fluctuations suggest input variability, which may affect model generalization. (b) Nova Scotia: High noise levels can disrupt learning patterns and reduce prediction accuracy. (c) Denmark: Mixed noise frequencies may lead to overfitting or obscure useful signals. (d) Italy: Smooth data with minimal noise helps models achieve better accuracy. (e) France: Sharp outliers may bias learning and increase prediction errors. (f) Spain: Load curve for Spain with noise points identified by red markers.

when comparing across datasets with widely varying scales, as it normalizes errors and allows us to assess performance on an even playing field—making it crucial in this multi-context evaluation. Fig. 2 reveals that each model demonstrates its best performance on different datasets.

Table 10
The variability of datasets.

Dataset	Interquartile range (IQR)
Portugal dataset	1524.25
Spain dataset	7211
France dataset	15,954.5
Italy dataset	13,067.5
Nova Scotia dataset	356.25
Denmark dataset	1266.25

CatBoost, GRU, and RNN achieved their lowest nRMSE % on the Spain dataset, showing strong adaptability to that data. LightGBM and XGBoost performed best on the Portugal dataset, both around 2.5 % nRMSE. TCN and LSTM showed better results on the Nova Scotia dataset, with LSTM reaching its minimum at 7.9 %. These results indicate that no single model consistently outperforms others across all datasets. A model that excels on one dataset may perform poorly on another, highlighting the importance of tailoring model selection to specific data characteristics rather than relying on general performance.

The noise detection in the datasets was determined based on a threshold applied to the absolute difference between each data point and the mean value of the dataset. A low threshold of 2 was chosen to ensure stricter noise detection and avoid missing potential noisy data points. Fig. 3 illustrates the distribution of noisy points in the datasets. While

Table 11
The models' accuracy across the datasets after increasing the dataset size.

Dataset	LightGBM of nRMSE %		XGBoost of nRMSE %		CatBoost of nRMSE %		TCN of nRMSE %		GRU of nRMSE %		RNN of nRMSE %		LSTM of nRMSE %	
	Before	After	Before	After	Before	After	Before	After	Before	After	Before	After	Before	After
Portugal	2.57	1.6	2.74	2.53	2.5	1.6	7.72	4.83	7.99	5.47	4.87	4.21	9.53	7.87
Nova Scotia	3.47	2.2	2.9	1.9	3.22	2.52	6.64	4.98	7.10	6.12	4.52	3.11	7.88	6.52
France	3.15	1.5	3.86	3.02	2.8	1.58	8.39	6.02	7.47	6.23	6.12	5.65	9.23	7.79
Italy	2.92	1.6	3.09	2.8	2.6	1.6	7.51	8.24	9.88	8.77	6.09	5.68	13.51	12.34
Denmark	3.28	1.86	3.44	2.7	2.91	1.7	7.2	8.2	9.10	8.3	6.28	6.20	9.5	7.9
Spain	2.31	**1.3**	2.83	2.22	2.37	1.37	6.69	5.27	6.29	5.79	4.8	**3.9**	10.05	8.89

Table 12
Comparative evaluation of forecasting models in terms of accuracy and model complexity.

Model	Accuracy	Efficiency (accuracy / complexity)	Recommended use
LightGBM	Best overall	High accuracy, low complexity	Most practical and scalable
CatBoost	Very good	High accuracy, moderate complexity	Alternative to LightGBM
XGBoost	Good	Decent accuracy, higher MAE	Use when interpretability is needed
RNN	Moderate	Balanced but outdated	Only for basic time series
GRU	Poor	Low R^2, high error	Not ideal
TCN	Poor	High complexity, weak accuracy	Not recommended
LSTM	Worst	High complexity, low accuracy	Avoid with current setup

noisy points are present in all datasets, there is no clear correlation between the presence of noise and the variance in model performance across the datasets. For instance, the Italy dataset exhibits fewer noisy points, yet the models achieved worse performance. The randomness of datasets has a direct impact on the performance of machine learning and deep learning models. Running Chi-Square and Runs Test showed that the datasets do not appear to be generated by a random process, and there are no systematic patterns or predictability.

Table 10 displays the variability of datasets, measured by the Interquartile Range (IQR). It is clear that the Italy dataset has the greatest variability, while the Nova Scotia dataset has the smallest variability.

The data analysis results did not show any distinct special behavior across all the datasets that could explain why the models' performance varied across the datasets.

To improve the performance of machine and deep learning models, the dataset size has been increased to three years.

Table 11 illustrates the models' accuracy across the datasets after increasing the dataset length to cover a two-year period. Overall, the performance of the models shows an increasing trend in the nRMSE values for most models. The LightGBM model achieved the highest accuracy of nRMSE with 1.3 across the Spain dataset, while the worst accuracy of nRMSE was observed across the LSTM model with 12.34. Although the TCN model for the Portugal dataset, RNN model for the Nova Scotia dataset, and LSTM model for the Italy dataset did not initially demonstrate noticeable accuracy, their nRMSE values notably decreased after three years.

The observation that the models performed differently on the Spain dataset can be attributed to several factors that influence model behavior. Specifically, the Spain dataset exhibits certain characteristics, such as seasonal variations, distinct consumption patterns, and potential outliers, that may affect the performance of the forecasting models. These patterns might be more challenging for some models to capture, especially deep learning models, which require extensive training data to effectively learn complex temporal dependencies. Moreover, the inherent noise and randomness in the data could contribute to varying model performance across different algorithms. Some models, such as LightGBM and CatBoost, which are gradient-boosting based and typically perform well in handling non-linearity and complex relationships, were able to generalize better and achieve higher accuracy on this particular dataset. In contrast, deep learning models like LSTM, which rely on sequential data and long-term dependencies, may struggle with such datasets if not properly tuned or if the model depth and complexity do not align well with the data's inherent structure.

Table 12 presents a comprehensive comparative analysis of forecasting performance and model complexity across the six models: LightGBM, XGBoost, CatBoost, TCN, GRU, RNN, and LSTM. The evaluation is based on the provided metrics across six datasets: Portugal, Nova Scotia, France, Italy, Denmark, and Spain.

When comparing the boxplots of prediction errors for the LSTM and LightGBM models, it's easy to see why LightGBM performs better in forecasting load consumption. Based on Table 3, we selected the best and worst-performing models, LightGBM and LSTM, respectively for a detailed comparison. Fig. 4 shows the errors from the LightGBM model are much more tightly grouped around the center, which means it makes more consistent and reliable predictions. In contrast, the LSTM model shows a much wider spread in its errors, indicating that its performance varies more significantly, Fig. 5. Even when we disregard the outliers, the core of the data (the box itself) clearly reveals that LightGBM not only stays closer to the actual values but also maintains more stable predictions. This tighter and more centered distribution of errors highlights LightGBM's superior ability in handling the forecasting task.

One of the key limitations of our study is the high computational cost of training deep learning models. Unlike machine learning models such as LightGBM, XGBoost, and CatBoost, deep learning models like GRU, RNN, TCN, and LSTM require significantly more resources for both training and hyperparameter tuning. This limitation affected our study in several ways:

- Training Time and Resource Constraints: Deep learning models took much longer to train due to their complexity and large number of parameters. This made it difficult to conduct extensive hyperparameter tuning, which could have improved their performance. In contrast, machine learning models—especially LightGBM—delivered high accuracy while requiring far less computational power.
- Scalability and Practical Deployment: While deep learning models have shown great potential in forecasting applications, their high computational demands make them less practical for real-time use or deployment in resource-limited environments. Our findings suggest that machine learning models, which are much more efficient, may be a better fit for large-scale load forecasting.
- Dataset Size and Model Performance: Increasing the dataset size (from one year to three years) improved the accuracy of all models, particularly the machine learning ones. However, the deep learning models didn't show as much improvement. This may be because optimizing their architectures requires even more computational resources, which we had to limit. With greater resources, additional tuning could have potentially improved their accuracy.

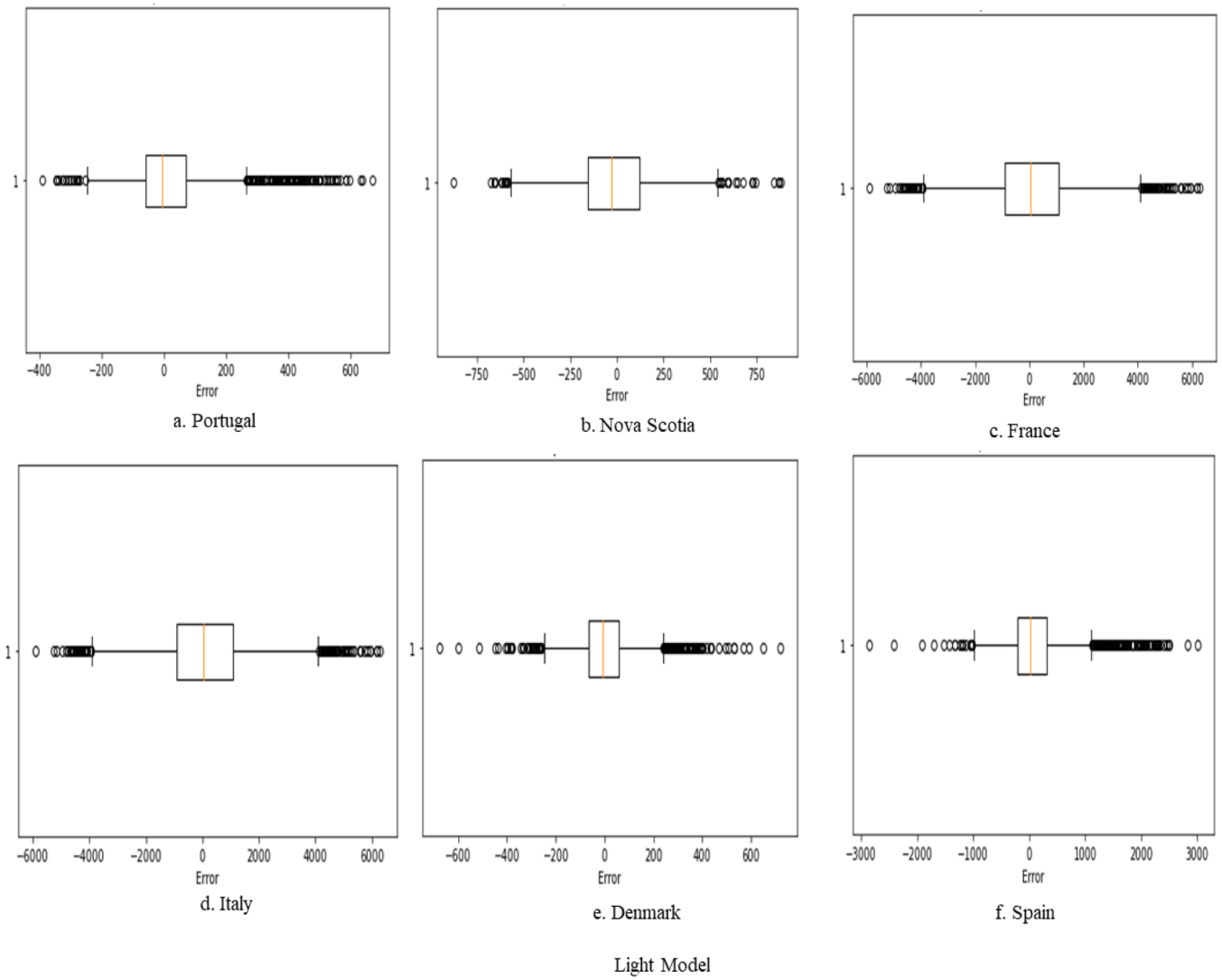


Fig. 4. Boxplots of prediction errors for the Light Model across different regions. (a) Portugal: The Light model shows relatively narrow error dispersion, with some positive outliers. (b) Nova Scotia: Errors are centered on zero, but a few large deviations are observed. (c) France: The Light model maintains a balanced error spread with moderate variability. (d) Italy: A wider error range is observed, indicating greater prediction variability. (e) Denmark: The Light model achieves a tight error distribution with few extreme outliers. (f) Spain: Errors are mostly centered, though some spread toward positive values is noticeable.

5. Conclusion

Forecasting electricity consumption with precision is essential for maintaining grid reliability and supporting informed decision-making. In this study, a variety of machine learning and deep learning models were assessed using hourly load data collected over a one-year period from multiple regions. The analysis highlighted noticeable differences in model accuracy across datasets, even when using the same hyper-parameter configurations. Among all models, LightGBM and CatBoost consistently offered the most reliable and accurate predictions, yielding the lowest error margins and highest R^2 scores. LightGBM delivered its best results on the Spain dataset, achieving an nRMSE of 2.31 %, MAE of 397.8, and R^2 of 0.98. Similarly, CatBoost performed strongly on the same dataset, with an nRMSE of 2.3 %, MAE of 407.9, and R^2 of 0.98. XGBoost also performed well, though with less consistency, reaching an nRMSE of 3.09 % and R^2 of 0.98 on the Italy dataset. On the other hand, deep learning models like LSTM, GRU, RNN, and TCN generally showed weaker results. The LSTM model, for instance, had a relatively high error on the Portugal dataset, with an nRMSE of 9.53 % and R^2 of only 0.60. Among the deep learning approaches, RNN done better than the rest, managing an nRMSE of 4.8 % and R^2 of 0.90 on the Spain dataset. However, models such as GRU and TCN struggled with larger errors and lower accuracy—TCN's performance on the France dataset, for example, included an nRMSE of 8.39 % and R^2 of 0.57.

While this study provides valuable insights into the performance of various models across different datasets, future work will focus on investigating advanced hybrid models that integrate both machine learning and deep learning approaches to improve forecasting accuracy. Additionally, incorporating explainable AI (XAI) methods may improve the interpretability and trustworthiness of the model outputs. We also recognize the importance of benchmarking against classical statistical models. Including baseline models such as ARIMA or naive forecasting could provide a meaningful reference point for evaluating the added value of more complex architectures. A comparative study involving these baselines will be considered in future investigations. Moreover, further analysis of external factors—such as weather conditions and economic indicators—may offer a more holistic understanding of load consumption patterns. Finally, we plan to explore alternative hyper-parameter optimization strategies, including Bayesian Optimization and Genetic Algorithms, to assess their potential in improving model performance and stability.

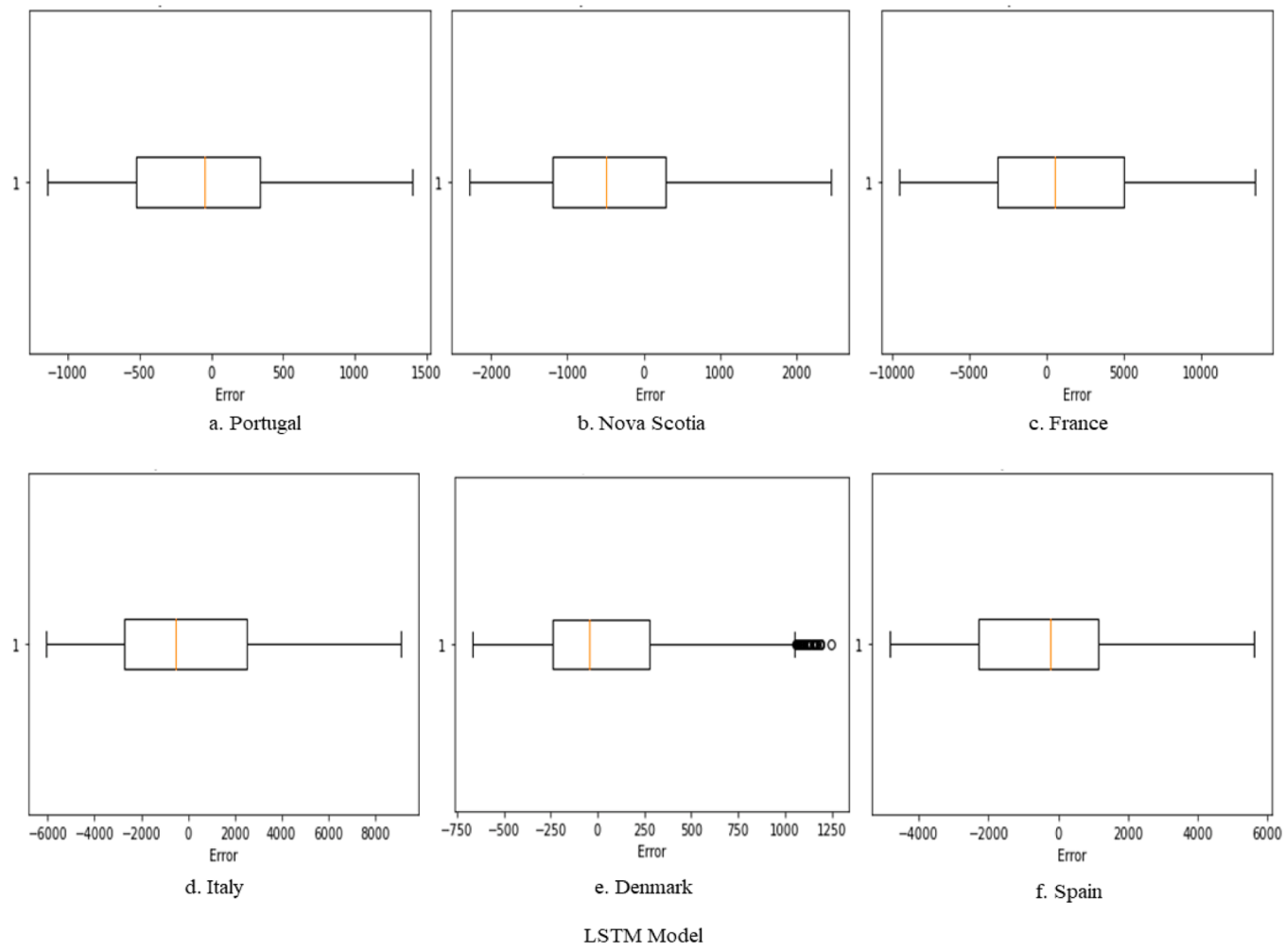


Fig. 5. Boxplots of prediction errors for the LSTM Model across different regions. (a)Portugal: LSTM predictions have a compact error distribution with minimal outliers. (b) Nova Scotia: Errors are close to zero with slightly more symmetrical distribution. (c) France: A wider error range, but fewer extreme deviations compared to the Light model. (d) Italy: The LSTM model shows high consistency, though with larger extreme errors. (e) Denmark: Errors are tightly packed around zero, indicating stable performance. (f) Spain: Good prediction stability is seen, with a few positive error outliers.

CRediT authorship contribution statement

Hmeda Musbah: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Meftah Elsaraiti:** Writing – review & editing, Methodology, Investigation.

Declaration of competing interest

We don't have conflict of interest or funding resources to declare.

Data availability

Data will be made available on request.

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